Name: Akshaya Sri
Reg No: 230701024

EXPERIMENT - 4

Hash Cracking Using John the Ripper

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.
Login to the system.

Step 2: Open Terminal and Verify Tool

Open terminal and check if John the Ripper is installed:

```
john --version
```

Step 3: Create Hash File

Create a file and store a password hash:

```
nano hash.txt
```

Step 4: Run Hash Cracking

```
john hash.txt
```

The tool attempts to crack the hash using default wordlists.

Step 5: View Cracked Password

```
john --show hash.txt
```

```
┌──(student㉿kali)-[~]
└─$ john
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100 O
MP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.

┌──(student㉿kali)-[~]
└─$ nano hash.txt

┌──(student㉿kali)-[~]
└─$ john hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password         (test)
1g 0:00:00:00 DONE 2/3 (2026-02-03 15:56) 33.33g/s 445666p/s 445666c/s 445666
C/s 123456..Herman1
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

┌──(student㉿kali)-[~]
└─$ john --show hash.txt
test:password

1 password hash cracked, 0 left
```

Result

Password hash cracking was successfully demonstrated, highlighting the need for strong passwords and secure hashing techniques.